ANDHRA UNIVERSITY COLLEGE OF ENGINEERING (A)

DEPARTMENT OF MECHANICAL ENGINEERING, VISAKHAPATNAM

HYDRAULIC MACHINES & THERMO-FLUIDS CELL

★ COMPREHENSIVE LECTURE NOTES & FORMULA HANDBOOK ★

# Fluid Mechanics & Hydraulic Machinery Comprehensive Lecture Notes

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| **Course Code:** | AU-MECH-2103 | **Academic Scheme:** | II B.Tech I Semester (AU Regulation) |
| **Verification:** | Verified Authentic Academic Copy | **System Ref:** | STUDY-BLUEPRINT-VERIFIED-2026 |

## DETAILED CURRICULAR MODULES & TOPICS

### Module 1: Fluid Statics & Manometry

Continuum concept, Viscosity and Newton's law, Hydrostatic forces on submerged plane and curved surfaces, Buoyancy and metacentric height.

### Module 2: Fluid Kinematics & Dynamics

Streamlines, streaklines, pathlines, Continuity equation in 3D, Euler's and Bernoulli's equations, Venturimeter and orifice meter coefficient of discharge.

### Module 3: Laminar & Turbulent Flow in Pipes

Reynolds experiment, Hagen-Poiseuille law, Darcy-Weisbach friction factor, Minor and major losses, Hydraulic gradient line (HGL) and TEL.

### Module 4: Boundary Layer Theory

Boundary layer thickness, displacement, momentum and energy thickness, Blasius solution, Separation of boundary layer and control methods.

### Module 5: Hydraulic Turbines & Pumps

Pelton wheel, Francis and Kaplan turbines, Unit quantities and specific speed, Centrifugal pumps, priming and cavitation phenomena.

## EVALUATION SCHEME & EXAMINATION BLUEPRINT

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| **Internal Assessment** | 30 Marks |
| **Semester Examination** | 70 Marks |
| **Prescribed Reference** | Fluid Mechanics by Frank M. White / Dr. R.K. Bansal |

| OFFICIAL SEAL | AUTHENTICATED DOCUMENT | CONTROLLER OF EXAMINATIONS |
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